TASK 17: Dashboard Audit, Data Provenance, and Enhancement Report

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh | **Roll Number:** 24102A2005

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Status: AUDITED, ENHANCED & VERIFIED (45/45 Unit/Integration Tests Passing, 6/6 Python Tests Passing)

Executive Summary

This report documents the comprehensive audit and enhancement of the ParkWatch public parking dashboard following the integration of the AI computer-vision module. In accordance with strict DevOps and academic safety rules:

1. Baseline operational metrics were measured from the running database without modifying or resetting `data/parkingdb.mv.db`.
2. The root cause of the **24-versus-25 parking locations discrepancy** was thoroughly investigated and identified: the persistent database originally seeded 24 pan-Indian metropolitan locations, and location #25 (`PPL-099 Civic Center Innovation Garage`) was dynamically created during the automated endpoint regression suite in Task 14 (`scripts/test-endpoints.ps1`). The database seed guard (`if (repo.count() > 0) return;`) rightly prevented subsequent overwrites.
3. A transparent **Data Provenance Audit** was performed: while source code in `SampleDataConfig.java` was previously drafted for 24 Mumbai BMC lots, the active persistent database actually serves an illustrative pan-Indian metropolitan sample (Mumbai, Delhi NCR, Bengaluru, Hyderabad, Chennai, Kolkata, Pune, Ahmedabad, Chandigarh). Real-time telemetry is simulated for academic demonstration.
4. Three high-value, safe improvements were implemented:

- **Fixed Hibernate Lazy Proxy JSON Serialization Error:** Eliminated an HTTP 500 failure on `GET /api/v1/events/recent` and `GET /api/v1/events/location/{id}` by adding `@JsonIgnoreProperties({"hibernateLazyInitializer", "handler"})`.
- **Global Event Audit Log View:** Created `@GetMapping("/events")` in `ParkingEventController` and enhanced `event-list.html` to provide an accessible audit log across all municipal facilities with direct navigation.
- **CSV / Data Export Endpoint:** Implemented `GET /api/v1/locations/export?format=csv` and integrated an **Export CSV** action on the main dashboard for administrative reporting.
- **Data Freshness Indicators:** Embedded real-time telemetry delay indicators (`>60 mins`) on the dashboard table rows.

1. Full regression test suites passed (45/45 Maven Spring Boot tests, 6/6 Python AI pipeline tests).

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1. Verified Baseline Operational Metrics

The application was inspected running on dedicated test port **8085** (preserving the active Jenkins CI service on port 8080). Operational metrics were queried directly from the REST API:

Metric	Measured Value	API Endpoint / Source
Total Parking Locations	`25`	`GET /api/v1/locations`
Total Parking Capacity	`8,040` spaces	`GET /api/v1/dashboard/summary`
Total Occupied Spaces	`4,531` spaces	`GET /api/v1/dashboard/summary`
Total Available Spaces	`3,509` spaces	`GET /api/v1/dashboard/summary`
Overall Utilization Rate	`56.4%`	`GET /api/v1/dashboard/summary`
Full Capacity Locations	`3` (`100%` saturation)	`GET /api/v1/locations?filter=FULL`
Low Availability Locations	`5` (`≥ 85%` utilization)	`GET /api/v1/locations?filter=LOW`
Available / Normal Locations	`17` (`< 85%` utilization)	`GET /api/v1/locations?filter=AVAILABLE`
Stale Telemetry Locations	`25` (`> 60 mins` old)	`GET /api/v1/locations?filter=STALE`
Total System Alerts	`33` (3 Full + 5 Low + 25 Stale)	`GET /api/v1/alerts`
Recent Parking Events	`1` recorded event in DB	`GET /api/v1/events/recent`

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2. Investigation: The 24-versus-25 Location Discrepancy

2.1 The Discrepancy

Earlier project documentation (Tasks 03, 04, and 15) reported **24** parking locations, whereas recent verification tasks reported **25** locations.

2.2 Root Cause Analysis

An inspection of the database records and git commit history revealed the exact chain of events:

1. **Initial Seeding (Task 03 / Commit `53f42f9d`):**

The application seeded 24 initial locations (`PPL-001` through `PPL-024`) into the persistent file-backed H2 database at `data/parkingdb.mv.db` .

1. **Automated Endpoint Testing (Task 14 / Commit `d40dae0a`):**

The test script `scripts/test-endpoints.ps1` lines 40–54 performed a functional test of the `POST /api/v1/locations` endpoint:

```powershell
$newLocBody = @{
name = "Civic Center Innovation Garage"
area = "Cyber City"
address = "500 Tech Boulevard"
totalCapacity = 100
occupiedSpaces = 25
locationCode = "PPL-099"
contactPhone = "+91 98765 43210"
description = "Smart automated municipal lot"
active = $true
```

} | ConvertTo-Json

$created = Invoke-RestMethod -Uri "http://localhost:8080/api/v1/locations" -Method Post -Body $newLocBody ...

```

This created **Location #25** ( `PPL-099 Civic Center Innovation Garage` ).

1. **Database Guard Preservation:**

In `SampleDataConfig.java` , line 35 enforces:

````java

if (repo.count() > 0) {

log.info("Parking location database already initialized ({} locations present). Skipping seed.", repo.count());

return;

}

```

Because `data/parkingdb.mv.db` is a persistent database file and strict project rules forbid deleting or resetting `parkingdb.mv.db` , the 25th record persisted across all subsequent test runs and restarts.

2.3 Verdict

The difference is **not** a database corruption or a bug; it is a **legitimate persistent test entity** created during end-to-end integration testing in Task 14. In adherence to the safety rule "*Do not delete or modify any records to make the numbers match*", the record is preserved as an active demonstration facility.

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3. Data Provenance Audit

An audit of the active records in `data/parkingdb.mv.db` establishes the true provenance of the dashboard dataset:
```

ID	Location Code	Facility Name	Geographic Locality	Metro Region	Provenance Classification
1	`PPL-001`	Bandra West Civic Plaza Parking	Bandra West, SV Road	Mumbai (MMR)	Illustrative Sample Data
2	`PPL-002`	Koramangala IT Park Parking	Koramangala 4th Block	Bengaluru	Illustrative Sample Data
3	`PPL-003`	Sector 17 Government Parking	Sector 17 Plaza	Chandigarh	Illustrative Sample Data
4	`PPL-004`	Connaught Place Underground Parking	Block A, Connaught Place	New Delhi (NCR)	Illustrative Sample Data
5	`PPL-005`	Jubilee Hills Club Parking	Road No. 36, Jubilee Hills	Hyderabad	Illustrative Sample Data
6	`PPL-006`	Vashi Railway Station Parking	Sector 17, Vashi	Navi Mumbai (MMR)	Illustrative Sample Data
7	`PPL-007`	MG Road Metro Parking	MG Road Station	Bengaluru	Illustrative Sample Data
8	`PPL-008`	Salt Lake Sector V Parking	Webel More, Sector V	Kolkata	Illustrative Sample Data
9	`PPL-009`	Garware College Road Parking	FC Road, Shivajinagar	Pune	Illustrative Sample Data
10	`PPL-010`	Anna Nagar Tower Parking	3rd Avenue, Anna Nagar West	Chennai	Illustrative Sample Data
11	`PPL-011`	Phoenix MarketCity Parking - Block C	Whitefield Main Road	Bengaluru	Commercial Retail Sample
12	`PPL-012`	Linking Road Shopping Parking	Linking Road, Bandra West	Mumbai (MMR)	Illustrative Sample Data
13	`PPL-013`	Laxmi Nagar District Centre Parking	District Centre, Laxmi Nagar	New Delhi (NCR)	Illustrative Sample Data
14	`PPL-014`	Hitech City Central Parking	Cyber Towers, Hitech City	Hyderabad	Commercial Tech Hub Sample
15	`PPL-015`	Law Garden Night Market Parking	Law Garden Road, Ellisbridge	Ahmedabad	Illustrative Sample Data
16	`PPL-016`	Sarojini Nagar Market Parking	Sarojini Nagar Market	New Delhi (NCR)	Municipal Market Sample
17	`PPL-017`	Dadar Station East Parking	Dadar East	Mumbai (MMR)	Railway Transit Sample
18	`PPL-018`	Commercial Street Parking	Commercial Street	Bengaluru	Commercial Retail Sample
19	`PPL-019`	Churchgate Station Parking	Marine Lines	Mumbai (MMR)	Railway Transit Sample
20	`PPL-020`	Karol Bagh Multi-Level Parking	Ajmal Khan Road	New Delhi (NCR)	Municipal Lot Sample
21	`PPL-021`	Egmore Bus Terminus Parking	EVR Periyar Salai	Chennai	Transit Hub Sample
22	`PPL-022`	Majestic Bus Stand Parking	Kempegowda Bus Station	Bengaluru	Transit Hub Sample
23	`PPL-023`	Sector 18 Noida Parking	Sector 18 Commercial Hub	Noida (NCR)	Commercial Center Sample
24	`PPL-024`	Vastrapur Lake Parking	Vastrapur Lake Road	Ahmedabad	Civic Amenity Sample
25	`PPL-099`	Civic Center Innovation Garage	500 Tech Boulevard, Cyber City	Test Facility	Dynamic Integration Test Entity

Provenance Audit Conclusions:

- **Pan-Indian Metropolitan Scope:** The active database does not contain purely BMC Mumbai municipal lots; it represents a diverse cross-section of 9 Indian urban hubs across 8 states/UTs.
- **Simulated Real-Time Feeds:** Brihanmumbai Municipal Corporation (BMC/MCGM) does not expose an open real-time sensor API for private third-party ingestion. Therefore, all occupancy numbers, timestamps, and sensor telemetry represent simulated operational data for DevOps testing and academic demonstration.

- **AI Separation:** AI computer-vision predictions from `POST /ai-analysis` (derived from the Brazilian PKLot dataset) remain strictly decoupled from this database and never alter municipal lot counts.

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4. Dashboard Feature Audit & Testing

Feature Under Audit	Tested Endpoint / Route	Verification Command / Method	Status	Audit Findings
Search Functionality	`GET /?search=Bandra` & `GET /api/v1/locations?search=Bandra`	`curl.exe -s "http://localhost:8085/?search=Bandra"`	**WORKING**	Correctly filters and returns 2 matching facilities (Bandra Civic Plaza, Linking Road).
Status Filtering	`GET /?filter=FULL` & `GET /api/v1/locations?filter=...`	REST API calls with `FULL`, `LOW`, `AVAILABLE`, `STALE`	**WORKING**	Correctly partitions locations into `FULL` (3), `LOW` (5), `AVAILABLE` (17), `STALE` (25).
Location Details	`GET /locations/1`	`Invoke-WebRequest -Uri 'http://localhost:8085/locations/1'`	**WORKING**	HTTP 200; displays capacity progress, contact phone, timestamps, and action buttons.
Add / Edit Facilities	`GET /locations/new` & `GET /locations/1/edit`	HTTP GET and validation tests	**WORKING**	HTTP 200; forms validate capacity ≥ 1, occupied $\le$ capacity, and required strings.
Occupancy Boundary Patch	`PATCH /api/v1/locations/1/occupancy`	Tested boundary payload (`occupiedSpaces > totalCapacity`)	**WORKING**	Correctly rejected with HTTP 400 Bad Request; valid occupancies update in place.
Recent Event History	`GET /api/v1/events/recent`	`curl.exe -s http://localhost:8085/api/v1/events/recent`	**FIXED**	Previously failed with HTTP 500 (Jackson lazy proxy bug). Now returns HTTP 200 with JSON array.
Global Audit Log UI	`GET /events`	`curl.exe -s http://localhost:8085/events`	**NEW**	Rendered global event audit log with facility links and operator notes.
Alerts & Exceptions	`GET /alerts` & `GET /api/v1/alerts`	`Invoke-RestMethod -Uri 'http://localhost:8085/api/v1/alerts'`	**WORKING**	HTTP 200; returns 33 alerts (3 Full, 5 Low Availability, 25 Stale Telemetry).
AI Stall Analysis	`GET /ai-analysis` & `POST /ai-analysis`	End-to-end inference using YOLOv8n checkpoint	**WORKING**	HTTP 200; 166 stalls detected, annotated JPEG served via `/ai-analysis/image/...`.
CSV Data Export	`GET /api/v1/locations/export?format=csv`	`curl.exe -s http://localhost:8085/api/v1/locations/export`	**NEW**	HTTP 200; exports all 25 facilities with full metadata in CSV format.

5. Improvements Implemented

Three safe, architectural improvements were introduced to strengthen the dashboard without altering the existing schema, database, or model:

1. Fixed Hibernate Lazy Proxy JSON Serialization

- **Issue:** Calling `GET /api/v1/events/recent` or `GET /api/v1/events/location/{id}` triggered `com.fasterxml.jackson.databind.exc.InvalidDefinitionException: No serializer found for class org.hibernate.proxy.pojo.bytebuddy.ByteBuddyInterceptor`.
- **Solution:** Added `@JsonIgnoreProperties({"hibernateLazyInitializer", "handler"})` to ParkingLocation.java and ParkingEvent.java.
- **Result:** Event REST APIs now serialize clean JSON records with nested location details and zero exceptions.

2. Global Event Audit Log & Navigation

- **Issue:** Users could record events via `/events/new` or view events for a single facility via `/events/location/{id}`, but there was no global audit view across all municipal lots.
- **Solution:** Added `@GetMapping` in ParkingEventController.java, enhanced event-list.html to support both global and per-facility listings, and added **Events Log** to the navigation bar in header.html.

3. Municipal Data Export (CSV) & Freshness Indicators

- **Issue:** No export capability existed for administrative reporting, and row-level data staleness lacked visual prominence.
- **Solution:**

1. Implemented `GET /api/v1/locations/export` in ParkingApiController.java with RFC 4180 CSV escaping and `attachment; filename="parkwatch_locations.csv"`.
2. Added an **Export CSV** button and direct link on dashboard.html.
3. Added data freshness warning badges (`bi-clock-history`) to the **Last Updated** column for rows exceeding the 60-minute threshold.

6. Verification and Regression Testing

6.1 Unit and Integration Test Results

Executed: `mvn test -Dtest="!Selenium"`

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.parking.dashboard.controller.AiAnalysisControllerTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 15.20 s
[INFO] Running com.parking.dashboard.controller.ParkingApiControllerTest
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.669 s
[INFO] Running com.parking.dashboard.controller.ParkingEventControllerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.202 s
[INFO] Running com.parking.dashboard.ParkingDashboardApplicationTests
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.544 s
[INFO] Running com.parking.dashboard.service.ParkingLocationServiceTest
[INFO] Tests run: 21, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.263 s
[INFO]
[INFO] Results:
[INFO] Tests run: 45, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
```

Total: 45 passed, 0 failures, 0 errors (6 new integration tests added for CSV export, recent event serialization, and global audit log rendering).

6.2 AI Pipeline Regression Tests

Executed: `python tests/test_ai_pipeline.py`

```
Ran 6 tests in 0.079s
OK
```

Total: 6 passed, 0 failures.

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7. Known Limitations and Transparency Disclosures

1. **Simulated Telemetry:** Sensor feeds and occupancy numbers remain simulated for academic demonstration. BMC does not provide an open public sensor API for real-time occupancy.
2. **Pilot AI Model:** The YOLOv8n computer vision model was trained for 1 epoch on 174 subsampled PKLot images ($mAP_{50} = 0.0063$, Precision = 0.0087, Recall = 0.0500) and is strictly designated as an experimental academic demonstration.
3. **Port 8080 Utilization:** Port 8080 is reserved by the Jenkins CI service (PID 7180) on the host machine. The dashboard application safely binds to configured ports (e.g. 8081 or 8085).

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8. Git Safety Check

Executed: `git status`

- **Staged files:** None (`no changes added to commit`).
- **Unstaged changes:** Only the audited and enhanced files.
- **Untracked files:** Only new report artifacts, documentation, and the new unit test file.
- **Safety Compliance:** Zero database files (*`.mv.db`*), *weights* (`.pt`), dataset images, or secrets were committed or pushed.